Ram Prakash Yallavula

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Education

University at Buffalo (SUNY) — Buffalo, NY
Master of Science in Engineering Science — *August 2024 – January 2026*

KL University — India
Bachelor of Technology in Computer Science (Honors) — *July 2020 – May 2024*

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, C++, SQL, Bash
Backend & APIs: FastAPI, Spring Boot, Django, REST APIs, JWT, OAuth2, OOP, MVC
Frontend: React, TypeScript, Streamlit
Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, GitHub Actions, CI/CD, Linux
Databases: PostgreSQL, MySQL
Tools & Practices: Git, Postman, Debugging, Unit Testing, SDLC, Agile/Scrum, Technical Documentation
Data & ML: Apache Spark, LightGBM, scikit-learn, Hopsworks, Feature Engineering, ETL Pipelines

Experience

Software Development Intern — March 2022 – May 2022
AICTE — *India*

- Automated repetitive development workflows using **2 programming languages**, Java and Python, applying modular design and established coding standards to produce reusable software components.
- Delivered REST API functionality across Agile sprint cycles, resolving integration issues and coordinating with technical and non-technical stakeholders to ship maintainable software changes.

Projects

TaskFlow AI | ***React, TypeScript, FastAPI, PostgreSQL, Docker, OpenAI API***

- Engineered a full-stack SaaS platform spanning **7 API domains** using FastAPI, React/TypeScript, and PostgreSQL with JWT authentication and user-scoped task and project management.
- Implemented **3 AI workflows** for natural-language task parsing, subtask generation, and daily schedule generation using the OpenAI SDK with mock and local fallback support.
- Developed analytics and scheduled reminder systems across **4 core data models**—users, tasks, projects, and notifications—with background processing and interactive dashboards.
- Containerized **3 application services**—React frontend, FastAPI backend, and PostgreSQL—and automated backend testing and frontend production builds through GitHub Actions CI.

Project Management System | ***Python, Django, PostgreSQL, AWS EC2, REST APIs***

- Architected and deployed a Django REST API backend on AWS EC2 with role-based access control, secure authentication, PostgreSQL persistence, and JSON-based client-server communication.
- Established automated GitHub Actions CI/CD deployments while implementing task tracking and deadline-management workflows with Django ORM.

CitiBike Demand Forecasting Platform | ***Python, LightGBM, Hopsworks, GitHub Actions*** | **Live**

- Created **3 production ML pipelines** for feature ingestion, model training, and inference, aggregating Citi Bike trip data into hourly station-level demand forecasts using Hopsworks and LightGBM.
- Productionized next-hour demand forecasting with model versioning and **3 reliability fallback paths** covering sparse data, timestamp drift, and transient backend failures while serving predictions through a live Streamlit dashboard.

NYC Taxi Data Processing Pipeline | ***Python, Apache Spark, AWS S3, ETL***

- Engineered a distributed Spark ETL pipeline on AWS S3, separating large-scale NYC taxi processing into **3 stages**—ingestion, transformation, and validation—for fault-tolerant execution.
- Designed cloud-based data workflows with modular Spark processing components to preserve compatibility and reliability across large-scale taxi pipeline environments.

Certifications

AWS Certified Cloud Practitioner | AWS Certified Solutions Architect – Associate | Red Hat Certified Enterprise Application Developer